

Over Voltage Protection using ESP32

Design approach

I have designed this project using ESP32 in WOKWI (online simulation platform). My designed system can detect over voltage and over current fault and take necessary step to protect the load connected to it. When the voltage or current is over the set point it will trip with in 500ms and if the voltage and current are in suitable value after 5000ms of delay the load will be automatically turn on. I don't have the current sensor so I decided to go for simulation instead of hardware implementation also my time is limited for this task. But it will be more dramatic if I could make it with hardware setup.

Implementation

I used **WOKWI** online platform and **esp32** module. WOKWI is a free online simulation platform so I decided to do this task in it. This project can be easily made by Arduino boards but I am used to esp32 recent days. In schematics I used two potentiometers which represent the output of **ZMPT101B**(voltage sensor) and **ACS712**(current sensor). I2C 16*2 LCD display for monitoring the status. Relay module is used to connect or disconnect the Load. LED indicate the fault voltage. The 3 button are used to set the over voltage, over current and delay values. Also, the limiting voltage and current can be adjusted by serial monitor (computer control). The current status can be seen by serial command.

Button Mode : By pressing Blue button – to go to set mode; Green and Yellow button are used as up and down operation.

Serial Command :

HELP : To see all the command.
STATUS : To see the current Situation.
SET : To set the over voltage or over current value. For example, SET OV 250; SET OC 6
DELAY : To set the delay time. (ms)
RECON : To set the reconnecting load time. (ms)

Results

Set Over Voltage	Set Over Current	Supply Voltage	Supply Current	Load Status
250 V	6 A	226.7 V	3.63 A	Connected
250 V	6 A	260.1 V	3.63 A	Disconnected
280 V	6 A	265.6 V	3.63 A	Connected
280 V	6 A	294.4 V	3.63 A	Disconnected

When the voltage is under the limited voltage/current the load remains on. When the voltage/current goes above the limited value the relay disconnects the load with in 500ms. After that if the voltage/current cold down it will take another 5000ms to turn on the load.

Limitations

Here over current can be also used as protection(tested) i have only shown over voltage protection. The under voltage protection also can be done by modifying the code. The real hardware setup result can be little different but the main goal will achieve in simulation. Though in hardware setup we have to calibrate the voltage sensor but in simulation i skipped this part and used potentiometers as representation of the output of voltage and current sensors. It will be very pleasure for me if I do the hardware setup and add the under voltage protection.